

IOT-BASED SMART WASTE MANAGEMENT SYSTEM USING ULTRASONIC SENSOR AND BLYNK

a micro project e-report Submitted by

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under the supervision of

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BONAFIDE CERTIFICATE

This is to certify that the micro project e-report titled **“IOT-BASED SMART WASTE MANAGEMENT SYSTEM USING ULTRASONIC SENSOR AND BLYNK”** is the bonafide work of **“SANTHOSA PRIYAN K A (URK25CS7100), MAHESHBABU K P (URK25CS7148), JOSH ANTONY K (URK25CS7084)”** who carried out the project work under my supervision.

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ABSTRACT

With the rapid growth of urbanization, efficient waste management has become a major challenge in maintaining clean and healthy environments. Traditional waste collection systems rely on manual monitoring, which often leads to overflowing bins, unpleasant surroundings, and increased risk of diseases. To address these issues, this project proposes an IoT-based Smart Waste Management System that automates the process of waste monitoring and disposal.

The system integrates an ultrasonic sensor to measure the level of waste inside the dustbin and an IR sensor to detect the presence of a user. When a person approaches the bin, the IR sensor triggers a servo motor to automatically open the lid, ensuring a completely contactless and hygienic operation. The ultrasonic sensor continuously monitors the fill level of the bin and converts the distance into a percentage value representing the bin capacity.

The collected data is transmitted to the Blynk IoT platform using WiFi, allowing real-time monitoring of the dustbin status through a mobile application. This enables users and authorities to track waste levels remotely and take timely action before overflow occurs. The system also improves efficiency by reducing the need for manual inspection and optimizing waste collection schedules.

This smart dustbin solution contributes to cleaner surroundings, better hygiene, and efficient waste management practices. It is highly suitable for deployment in smart cities, public areas, institutions, and residential complexes. Furthermore, the project can be enhanced by integrating features such as GPS tracking, alert notifications, and AI-based waste segregation for future development.

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INTRODUCTION

1.1 Background of the Problem

Waste management is a major concern in urban areas due to increasing population and consumption. Traditional waste collection methods involve periodic manual checking of bins, which is inefficient and time-consuming. Overflowing bins not only create an unpleasant environment but also attract pests and spread diseases.

There is a growing need for smart systems that can automate waste monitoring and improve efficiency. The integration of IoT technology in waste management provides a solution by enabling real-time monitoring and automation.

1.2 Industry Relevance

Smart waste management systems are widely used in:

- Smart cities and urban infrastructure
- Municipal waste management
- Hospitals and healthcare facilities
- Educational institutions
- Industrial environments

These systems help reduce operational costs and improve efficiency.

1.3 Aim and Objectives

Aim

To develop an IoT-based smart dustbin system that automates waste monitoring and improves hygiene.

Objectives

- To design an automatic lid-opening mechanism

- To measure waste levels using sensors
 - To provide real-time monitoring using IoT
 - To reduce human interaction with waste
 - To improve efficiency in waste collection
-

1.4 SDG Link

This project supports:

- **SDG 11: Sustainable Cities and Communities**
 - Promotes cleanliness and efficient waste management
-

2. LITERATURE REVIEW

Several studies have explored the use of IoT in waste management systems. Smart bins equipped with sensors have been developed to monitor waste levels and optimize collection schedules.

Ultrasonic sensors are widely used due to their accuracy in distance measurement. IR sensors are commonly used for object detection. Integration with cloud platforms enables real-time monitoring and data analysis.

Previous research shows that IoT-based systems significantly reduce overflow incidents and improve sanitation standards. However, many systems lack automation for lid operation, which this project addresses.

3. METHODOLOGY

3.1 System Overview

The system combines sensors, microcontroller, and IoT platform to automate waste management.

3.2 Components Used

Hardware Components

- Arduino board with WiFi capability
- Ultrasonic Sensor (HC-SR04)

- IR Sensor
- Servo Motor
- Connecting wires and power supply

Software Components

- Arduino IDE
 - Blynk IoT Platform
-

3.3 Working Principle

The working of the system is divided into two main parts:

1. Lid Automation

- IR sensor detects the presence of a user
- Signal is sent to Arduino
- Servo motor rotates to open lid
- After a delay, lid closes automatically

2. Waste Level Monitoring

- Ultrasonic sensor measures distance
 - Distance is converted into bin level percentage
 - Data is sent to Blynk app via WiFi
-

3.4 Algorithm

1. Start the system
 2. Initialize sensors and WiFi
 3. Read IR sensor value
 4. If object detected → open lid
 5. Measure bin level using ultrasonic sensor
 6. Send data to Blynk
 7. Close lid after delay
 8. Repeat process
-

3.5 Flowchart (Write in Word)

- Start
 - Sensor detection
 - Lid open/close
 - Measure level
 - Send data
 - Repeat
-

4. PROGRAM / CODE EXPLANATION

4.1 Overview

```
#define BLYNK_TEMPLATE_ID "Template ID"
#define BLYNK_TEMPLATE_NAME "Template Name"
#define BLYNK_AUTH_TOKEN "Auth Token"
```

```
#define BLYNK_PRINT Serial
```

```
#include <WiFiS3.h>
#include <BlynkSimpleWifi.h>
#include <Servo.h>
```

```
char ssid[] = "";
char pass[] = "12345678";
char auth[] = IGrcP6X5sb_ZKpn_rqoO7v6PEVMal7vb;
```

```
BlynkTimer timer;
```

```
#define echoPin 5
#define trigPin 6
#define irPin 7
```

```
Servo servo;
```

```
long duration;
int distance;
int binLevel = 0;
```

```
void ultrasonic()
{
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
```



```

digitalWrite(trigPin, HIGH);
delayMicroseconds(10);

digitalWrite(trigPin, LOW);

duration = pulseIn(echoPin, HIGH);
distance = duration * 0.034 / 2;

binLevel = map(distance, 21, 0, 0, 100);

Blynk.virtualWrite(V0, distance);
Blynk.virtualWrite(V1, binLevel);
}

void SMESensor()
{
  int ir = digitalRead(irPin);

  if (ir == HIGH)
  {
    servo.write(90);

    for (int i = 0; i < 20; i++)
    {
      ultrasonic();
      Blynk.virtualWrite(V2, 90);
      delay(100);
    }

    servo.write(0);
    Blynk.virtualWrite(V2, 0);
  }
  else
  {
    ultrasonic();
    delay(200);
  }
}

void setup()
{
  Serial.begin(9600);

  servo.attach(9);

  pinMode(irPin, INPUT);
}

```

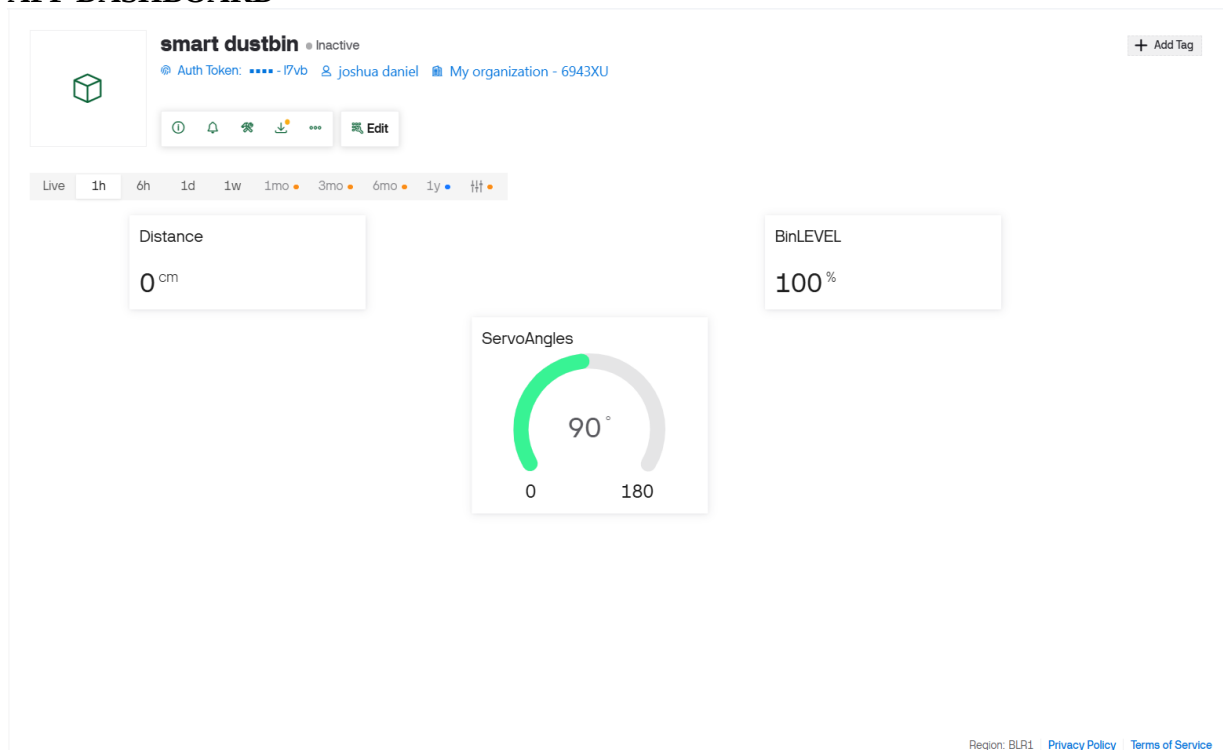
```
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);

Blynk.begin(auth, ssid, pass);

timer.setInterval(1000L, SMESensor);
}
```

```
void loop()
{
  Blynk.run();
  timer.run();
}
```

APP DASHBOARD



4.2 Key Functions

Ultrasonic Function

- Generates trigger pulse
- Measures echo time
- Calculates distance
- Converts to bin level percentage

Sensor Function

- Reads IR sensor input
- Controls servo motor
- Updates Blynk values

Setup Function

- Initializes pins
- Connects to WiFi
- Starts Blynk

Loop Function

- Runs continuously
 - Updates sensor data
-

4.3 Code Features

- Real-time data transmission
 - Automated control system
 - Efficient sensor integration
-

5. RESULTS AND INTERPRETATION

5.1 Results / Observations

- Automatic lid opening works correctly
 - Bin level displayed accurately in Blynk
 - System responds quickly to input
 - Contactless operation achieved
 - Reliable performance
-

5.2 Interpretation

The system demonstrates effective automation and monitoring. It reduces human effort and improves hygiene. The IoT integration ensures real-time data access.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The project successfully implements an IoT-based smart dustbin system. It improves waste management efficiency and promotes hygiene. The system is reliable, cost-effective, and suitable for real-world applications.

6.2 Future Scope

- GPS-based tracking system
 - SMS/notification alerts
 - Solar-powered system
 - AI-based waste segregation
 - Integration with municipal systems
-

7. ADVANTAGES

- Hygienic and contactless
 - Reduces manual effort
 - Real-time monitoring
 - Prevents overflow
 - Cost-effective
-

8. APPLICATIONS

- Smart cities
- Public places
- Hospitals
- Schools and colleges
- Offices and industries

9. LIMITATIONS

- Requires stable internet
- Sensor accuracy limitations
- Initial cost of setup

DECLARATION

I understand that Karunya Institute of Technology and Sciences retains the rights to any product developed (*software or hardware*) through the Micro Project submitted by me to the Institution.

Furthermore, if I intend to document or publish any findings from the Micro Project work for commercial purposes, I shall obtain a No Objection Certificate (NOC) from the Office of the Registrar before engaging in such activities.

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